$$z = t,$$

$$7y - 8t = 0 \;\Rightarrow\; y = \frac{8t}{7},$$

$$x + 3\left(\frac{8t}{7}\right) - 2t = 0 \;\Rightarrow\; x + \frac{24t - 14t}{7} = 0 \;\Rightarrow\; x = -\frac{10t}{7}.$$

So, the solution is $\left(x = -\frac{10t}{7}, y = \frac{8t}{7}, z = t\right)$, where t is any real number.

Example 1.37

Solve the system: $x + y - 2z = 0, 2x - 3y + z = 0, 3x - 7y + 10z = 0, 6x - 9y + 10z = 0.$

Solution

Here the number of equations is 4 and the number of unknowns is 3. Reducing the augmented matrix to echelon-form, we get

$$[A \mid O] = \left[\begin{array}{ccc|c} 1 & 1 & -2 & 0 \\ 2 & -3 & 1 & 0 \\ 3 & -7 & 10 & 0 \\ 6 & -9 & 10 & 0 \end{array}\right] \xrightarrow[R_4 \to R_4 - 6R_1]{\substack{R_2 \to R_2 - 2R_1, \\ R_3 \to R_3 - 3R_1,}} \left[\begin{array}{ccc|c} 1 & 1 & -2 & 0 \\ 0 & -5 & 5 & 0 \\ 0 & -10 & 16 & 0 \\ 0 & -15 & 22 & 0 \end{array}\right] \xrightarrow[R_3 \to R_3 \div (-2)]{\substack{R_2 \to R_2 \div (-5), \\ }} \left[\begin{array}{ccc|c} 1 & 1 & -2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 5 & -8 & 0 \\ 0 & -15 & 22 & 0 \end{array}\right]$$

$$\xrightarrow[R_4 \to R_4 + 15R_2]{R_3 \to R_3 - 5R_2,} \left[\begin{array}{ccc|c} 1 & 1 & -2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 7 & 0 \end{array}\right] \xrightarrow[R_4 \to R_4 \div 7]{R_3 \to R_3 \div (-3),} \left[\begin{array}{ccc|c} 1 & 1 & -2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right] \xrightarrow{R_4 \to R_4 - R_3} \left[\begin{array}{ccc|c} 1 & 1 & -2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

So, $\rho(A) = \rho([A \mid O]) = 3 =$ Number of unknowns.

Hence the system has trivial solution only.

Example 1.38

Determine the values of λ for which the following system of equations
$(3\lambda - 8)x + 3y + 3z = 0,\ 3x + (3\lambda - 8)y + 3z = 0,\ 3x + 3y + (3\lambda - 8)z = 0$

has a non-trivial solution.

Solution

Here the number of unknowns is 3. So, if the system is consistent and has a non-trivial solution, then the rank of the coefficient matrix is equal to the rank of the augmented matrix and is less than 3. So the determinant of the coefficient matrix should be 0.

Hence we get

$$\begin{vmatrix} 3\lambda - 8 & 3 & 3 \\ 3 & 3\lambda - 8 & 3 \\ 3 & 3 & 3\lambda - 8 \end{vmatrix} = 0 \text{ or } \begin{vmatrix} 3\lambda - 2 & 3\lambda - 2 & 3\lambda - 2 \\ 3 & 3\lambda - 8 & 3 \\ 3 & 3 & 3\lambda - 8 \end{vmatrix} = 0 \text{ (by applying } R_1 \to R_1 + R_2 + R_3)$$

$$\text{or } (3\lambda - 2)\begin{vmatrix} 1 & 1 & 1 \\ 3 & 3\lambda - 8 & 3 \\ 3 & 3 & 3\lambda - 8 \end{vmatrix} = 0 \text{ (by taking out } (3\lambda - 2) \text{ from } R_1)$$

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